California Housing Price Prediction using Linear Regression

Project Overview

This project analyzes the California Housing dataset using Python and Machine Learning. The objective is to perform data cleaning, exploratory data analysis (EDA), build Simple and Multiple Linear Regression models, evaluate their performance, and visualize the results using various statistical charts.

Dataset

Dataset Name: California Housing Dataset

File Used: `housing.csv`

The dataset contains information about California housing, including:

- Longitude
- Latitude
- Housing Median Age
- Total Rooms
- Total Bedrooms
- Population
- Households
- Median Income
- Median House Value (Target Variable)
- Ocean Proximity

Project Objectives

- Import and inspect the dataset
- Clean missing values and duplicate records
- Perform Exploratory Data Analysis (EDA)
- Calculate descriptive statistics (Mean, Median, Mode, Variance, Standard Deviation, IQR)
- Create professional visualizations
- Build a Simple Linear Regression model
- Build a Multiple Linear Regression model
- Evaluate model performance using R^2 Score, MAE, MSE, and RMSE
- Compare regression models
- Draw conclusions from the analysis

Technologies Used

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Plotly (Bonus)

Required Libraries

Install the required libraries using:

```
pip install pandas numpy matplotlib seaborn scikit-learn plotly
```

How to Run the Project

1. Install Python (3.x).
2. Install Jupyter Notebook.
3. Install all required libraries.
4. Place `housing.csv` in the same folder as the notebook.
5. Open the notebook in Jupyter.
6. Run all cells from top to bottom.

Machine Learning Models

Simple Linear Regression

- Feature Selected: `median_income`
- Target Variable: `median_house_value`

Evaluation Metrics:

- R^2 Score
- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)

Multiple Linear Regression

Used multiple important features to improve prediction accuracy.

Evaluation Metrics:

- R^2 Score
- MAE
- MSE
- RMSE

Performance was compared with the Simple Linear Regression model.

Visualizations Included

- Histograms
- Box Plots
- Scatter Plots
- Correlation Heatmap
- Regression Line Plot
- Residual Plot
- Interactive Plotly Visualizations (Bonus)

Key Findings

1. Median Income is the strongest predictor of house prices.
2. The dataset contained missing values only in the `total_bedrooms` column, which were handled during data cleaning.
3. House prices generally increase with higher median income.
4. Multiple Linear Regression produced better prediction accuracy than Simple Linear Regression.
5. Correlation analysis helped identify the most important features affecting house prices.
6. Visualizations made it easier to understand data distribution, relationships, and outliers.

Project Structure

```
California Housing Project/
│
├── California_Housing_Assignment.ipynb
├── housing.csv
├── README.md
```

```
├── requirements.txt
└── Screenshots/
```

Conclusion

This project demonstrates the complete workflow of a Machine Learning regression problem, including data preprocessing, exploratory data analysis, feature selection, model building, evaluation, and visualization. It provides practical experience with Python data science libraries and illustrates how statistical analysis supports predictive modeling.

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